US Patent & Trademark Office
Patent Public Search | Text View

United States Patent	12413576
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Bachorik; Justin et al.

Time-based token trust depreciation

Abstract

Disclosed herein are system, method, and device embodiments for time-based trust token (TBTT) depreciation. In an example embodiment, a service provider system (e.g., a service provider and API service) may receive a connection request including a demographic attribute associated with a first client account from a partner device, match the demographic attribute to client information associated with the first client account, send the partner device a connection request identifier and a URL including a depreciating token, and authenticate a second client account via a login page associated with the URL. Further, the service provider system may receive a verification request including the connection request identifier and the depreciating token, determine a security context of the depreciating token based on a depreciation function and the verification request, and determine, based on the security context, whether to create a connection between the second client account and partner device within the service provider system.

Inventors: **Bachorik; Justin (Silver Spring, MD), Randall; Randall (New York, NY), Shin; Brandee (Herndon, VA), Gray; Rocky (Fairfax, VA)**

Applicant: **Capital One Services, LLC** (McLean, VA)

Family ID: **75491552**

Assignee: **Capital One Services, LLC (McLean, VA)**

Appl. No.: **18/218147**

Filed: **July 05, 2023**

Prior Publication Data

Document Identifier	Publication Date
US 20240007457 A1	Jan. 04, 2024

Related U.S. Application Data

continuation parent-doc US 17476664 20210916 US 11743250 child-doc US 18218147
continuation parent-doc US 16654856 20191016 US 11171945 20201109 child-doc US 17476664

Publication Classification

Int. Cl.: **H04L9/40** (20220101); **G06Q20/10** (20120101); **G06Q20/38** (20120101)

U.S. Cl.:

CPC **H04L63/083** (20130101); **G06Q20/10** (20130101); **G06Q20/3821** (20130101);

Field of Classification Search

USPC: None

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6691232	12/2003	Wood et al.	N/A	N/A
9608974	12/2016	Barrows et al.	N/A	N/A
9836594	12/2016	Zhang et al.	N/A	N/A
9996835	12/2017	Dill et al.	N/A	N/A
10164999	12/2017	Casillas et al.	N/A	N/A
10708054	12/2019	Donaldson	N/A	H04L 9/3247
11171945	12/2020	Bachorik et al.	N/A	N/A
11216539	12/2021	Chang	N/A	H04L 67/562
2010/0205053	12/2009	Shuster	N/A	N/A
2012/0166803	12/2011	Hu et al.	N/A	N/A
2014/0279556	12/2013	Priebatsch	N/A	N/A
2016/0086184	12/2015	Carpenter et al.	N/A	N/A
2016/0087957	12/2015	Shah et al.	N/A	N/A
2016/0189119	12/2015	Bowman et al.	N/A	N/A
2016/0328707	12/2015	Wagner et al.	N/A	N/A
2017/0004506	12/2016	Steinman et al.	N/A	N/A
2017/0237726	12/2016	Wang	N/A	N/A
2017/0289168	12/2016	Bar	N/A	H04L 63/1408
2018/0075231	12/2017	Subramanian	N/A	H04L 63/0807
2018/0165781	12/2017	Rodriguez et al.	N/A	N/A
2019/0164156	12/2018	Lindemann	N/A	N/A
2019/0199723	12/2018	Tak et al.	N/A	N/A
2019/0229900	12/2018	Khristi	N/A	H04L 9/0827
2019/0372962	12/2018	Maria et al.	N/A	N/A
2019/0392428	12/2018	Bolla	N/A	N/A
2020/0137109	12/2019	Endler	N/A	N/A
2021/0119986	12/2020	Bachorik et al.	N/A	N/A

OTHER PUBLICATIONS

Author unknown, Enabling URL token validation, Jan. 2019, Fastly Help Guides, retrieved from https://docs.fastly.com/en/guides/enabling-url-token-validation (Year: 2019). cited by examiner

Primary Examiner: Cunningham, II; Gregory S

Attorney, Agent or Firm: Sterne, Kessler, Goldstein & Fox P.L.L.C.

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 17/476,664, filed on Sep. 16, 2021, which is a continuation of U.S. patent application Ser. No. 16/654,856, filed on Oct. 16, 2019, the contents of which is incorporated in its entirety herein.

BACKGROUND
(1) Many computing systems employ security tokens during authentication and/or authorization workflows. Typically, a security token is issued to a device and redeemable to access a digital resource. However, once a security token is issued, it may be captured by a malicious party and used in an attack (e.g., session hijacking) against a computing system. In addition, computing systems often fail to utilize security information provided by expired and/or stale security tokens during security workflows.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The accompanying drawings, which are incorporated herein and form part of the specification, illustrate the present disclosure and, together with the description, further serve to explain the principles of the disclosure and enable a person skilled in the relevant art to make and use the disclosure.
(2) FIG. **1** is a block diagram of an example framework for implementing time-based token trust depreciation, according to embodiments of the present disclosure.
(3) FIG. **2** is a sequence diagram illustrating an example process using time-based token trust depreciation, according to embodiments of the present disclosure.
(4) FIG. **3** is a sequence diagram illustrating an example process using time-based token trust depreciation, according to embodiments of the present disclosure.
(5) FIG. **4** is a sequence diagram illustrating an example process using time-based token trust depreciation, according to embodiments of the present disclosure.
(6) FIG. **5** is a flowchart illustrating a process implemented by a system for implementing time-based trust token depreciation, according to embodiments of the present disclosure.
(7) FIG. **6** illustrates a computer system, according to example embodiments of the present disclosure.
(8) The drawing in which an element first appears is typically indicated by the leftmost digit or digits in the corresponding reference number.
DETAILED DESCRIPTION OF THE INVENTION
(9) Provided herein are system, apparatus, device, method and/or computer program product embodiments, and/or combinations (including sub-combinations) thereof, for implementing time-based trust token (TBTT) depreciation.
(10) As mentioned above, security tokens may be used to perform malicious attacks (e.g., session

hijacking) that provide unauthorized access to computing resources. Embodiments herein address these challenges by employing a TBTT with a dynamic security context based on a trust depreciation function that limits opportunity for malicious use. Further, embodiments herein may further securely utilize a depreciated TBTT when providing access to a computing resource.
(11) FIG. **1** illustrates a diagram of an example framework for implementing a system for TBTT depreciation, according to embodiments of the present disclosure. As illustrated in FIG. **1**, a computing system **100** may include a service provider **102** including a task management application **104**, an API service **106**, a plurality of client devices **108**(**1**)-(N) associated with a plurality of clients **110**(**1**)-(N), a plurality of partner services **112**(**1**)-(N), and a partner platform **114**. Further, as illustrated in FIG. **1**, the service provider **102**, the API service **106**, the client devices **108**(**1**)-(N), the partner services **112**(**1**)-(N), and the partner platform **114** may communicate via a communication network(s) **116**. The communication network(s) **116** may include any or all of a private network, personal area network (PAN), Local-Area Network (LAN), Wide-Area Network (WAN), the Internet, or any other suitable network. Further, the connection between the service provider **102**, the API service **106**, the client devices **108**(**1**)-(N), the partner services **112**(**1**)-(N), or the partner platform **114**, and the communication network(s) **116** may be a wireless connection (e.g., Bluetooth, Wi-Fi connection, etc.), or a wired connection (e.g., Ethernet, universal serial bus (USB), etc.), or a combination thereof.
(12) In some embodiments, each client device **108** is associated with a client **110**. For instance, the client **110**(**1**) may employ the client device **108**(**1**), the client **110**(**2**) may employ the client device **108**(**2**), the client **110**(N) may employ the client device **108**(N), and so forth. Some examples of the client devices **108**(**1**)-(N) include, but are not limited to, tablets, smartphones, mobile devices, personal computers, laptop computers, appliances, IoT devices, wearables, etc. Further, as described in detail herein, the clients **110**(**1**)-(N) may employ the client devices **108**(**1**)-(N) to access resources provided by the service provider **102** or the partner services **112**(**1**)-(N) via the communication network **116**. In some embodiments, the partner platform **114** may be a centralized proxy or aggregator for the partner services **112**(**1**)-(N). Additionally, the API service **106** may be a web service providing one or more methods for implementing TBTT depreciation within the computing system **100**. For example, the service provider **102**, the client devices **108**(**1**)-(N), the partner services **112**(**1**)-(N), and the partner platform **114** may call the methods of the API service **106** to leverage TBTTs **118** for providing resources within the computing system **100**. In some examples, the API service **106** may be a simple object access protocol application programming interface (SOAP API) or a representational state transfer application programming interface (REST API)
(13) In some embodiments, the service provider **102** may be operated by or otherwise associated with a financial institution. In addition, the clients **110**(**1**)-(N) may have financial accounts with the financial institution. Further, the service provider **102** may be an online banking platform including the task management application **104**. As used herein, in some embodiments, “task management” may refer to creation, modification, deletion, performance, confirmation, administration, and completion of a task. As used herein, and in some embodiments, an “application” may refer to any computer program or software (e.g., client, agent, application, mobile application, web application, hybrid application, computer program, desktop application, or module) operable to run on the service provider **102** and/or client devices **108**(**1**)-(N). In addition, the task management application **104** may generate service tasks **120**(**1**)-(N) for the clients **110**(**1**)-(N), display the service tasks **120**(**1**)-(N) to the clients **110**(**1**)-(N) via the client devices **108**(**1**)-(N), and enable the clients **110**(**1**)-(N) to complete the service tasks **120**(**1**)-(N) via the client devices **108**(**1**)-(N). In some examples, a service task **120** may include an electronic bill or payment request from a first user of the task management application **104** to a second user of the task management application **104** for services rendered by the first user to the second user. For instance, the partner service **112**(**1**) may render a service to the client **110**(**1**). Further, the partner service **112**(**1**) may generate a service task

120(**1**) representing an electronic bill for the service rendered to the client **110**(**1**), and request that the task management application **104** display the service task **120**(**1**) to the client **110**(**1**) via the client device **108**(**1**). Additionally, the client **110**(**1**) may use a financial account accessible via the service provider **102** to pay the bill associated with the service task **120**(**1**). Some other examples of service tasks include billing notification, billing payment confirmation, document signature, information requests, etc.

(14) In some embodiments, the service provider **102** may require the partner service **112**(**1**) to have a connection to the client **110**(**1**) before permitting the partner service **112**(**1**) to assign the service tasks **120**(**1**)-(N) to the client **110**(**1**). Further, the service provider **102** may employ the TBTT **118**(**1**) to create the connection between the partner service **112**(**1**) and the client **110**(**1**). Although embodiments herein disclose employing the TBTTs **118** to access the task management application **104**, the TBTTs **118** may be implemented in other computing environments for authentication, authorization, or any other security process.

(15) As an example, the client device **108**(**1**) may send enrollment information **122**(**1**) to the partner service **112**(**1**). The enrollment information **122**(**1**) may represent consent by the client **110**(**1**) to connect with the partner service **112**(**1**) within the task management application **104**. For instance, the client device **108**(**1**) may send the enrollment information **122**(**1**) to the partner service **112**(**1**) based on the client **110**(**1**) selecting graphical controls (e.g., a checkbox and a button) indicating that the client **110**(**1**) intends to create the connection with the partner service **112**(**1**) via the service provider **102**. In response to receipt of the enrollment information **122**(**1**), the partner service **112**(**1**) may send a connection request **124**(**1**) to the API service **106**. The connection request **124**(**1**) may identify the partner service **112**(**1**) and the client **110**(**1**). In some embodiments, the connection request **124**(**1**) may include a partner service identifier identifying the partner service **112**(**1**), a client identifier identifying the client **110**(**1**), and one or more demographic attributes **126** associated with the client **110**(**1**). Some examples of a demographic attribute include a name, email address, telephone number, physical addresses, etc.

(16) Upon receipt of the connection request **124**(**1**), the API service **106** may determine whether the demographic attribute **126** matches client information **128**(**1**) associated with the client **110**(**1**). If the demographic attribute **126** matches the client information **128**(**1**), the API service **106** may generate the TBTT **118**(**1**) for creating the connection between the client **110**(**1**) and the partner service **112**(**1**) within the task management application **104**. The API service **106** may further store token information **130** including a connection request identifier **132**(**1**) identifying the request for connection between the client **110**(**1**) and the partner service **112**(**1**). In some embodiments, the TBTT **118** may include temporal information identifying the date and time of creation of the TBTT **118**(**1**). Additionally, or alternatively, the token information **130** may include temporal information identifying the date and time of creation of the TBTT **118**(**1**).

(17) Further, the API service **106** may send the connection information **134**(**1**) to the partner service **112**(**1**). The connection information **134**(**1**) may include a web application identifier **136**(**1**) including the TBTT **118**, and the connection request identifier **132**(**1**). In some examples, the web application identifier **136**(**1**) may be a universal resource locator (URL) with the TBTT **118** appended thereto within a query string. Further, the URL may be the address of a login page or other form of authentication for the task management application **104**. As referred to herein, in some embodiments, a “query string” may be a part of a URL which assigns values to specified parameters. In some embodiments, a query string includes fields added to a base URL by a Web browser or other client application, for example as part of an HTML, form.

(18) In response to receiving the connection information **134**(**1**), the partner service **112**(**1**) may redirect the client device **108**(**1**) to the login page represented by the web application identifier **136**(**1**). In some other embodiments, the partner service **112**(**1**) may embed the login page within a web page of the partner service **112**(**1**). After the redirection is performed, the client **110**(**1**) may perform an authentication process with the service provider **102** via the login page. In some

embodiments, the client device **108**(**1**) may send the login information **138** including the TBTT **118**(**1**) to the service provider **102** during the authentication process. Performing the authentication process ensures that the client device **108**(**1**) is associated with a client **110** having an account with the service provider **102**. But the TBTT **118**(**1**) is still required to provide proof that the API service **106** should create a connection between the client **110**(**1**) and the partner service **112**(**1**). For example, a malicious client **110**(**2**) of the service provider **102** may endeavor to create a connection between their account at the service provider **102** and the account of the client **110**(**1**) with the partner service **112**(**1**).

(19) As used herein, in some embodiments, "authentication" may refer to a process by which an entity (e.g., a user, a device, or a process) makes their identity known to a computing system (e.g., the service provider **102**). Further, authentication may be based on at least one of information an entity knows (e.g., a password, a PIN code, a digital certificate, a secret, a cryptographic key, a social security number, a zip code, a phone number, etc.), an object an entity possesses (e.g., a smart card, a hardware token, a dynamic password generator, etc.), or something an entity is (e.g., a fingerprint, a voiceprint, a retinal scan, etc.).

(20) In some embodiments, the service provider **102** may determine that the client **110**(**1**) endeavors to create a connection at the service provider **102** between the client **110**(**1**) and the partner service **112**(**1**) based upon the redemption of the TBTT **118**(**1**) via the authentication process. Additionally, the service provider **102** may request that the API service **106** verify the TBTT **118**(**1**). For example, the service provider **102** may send a verification request **140**(**1**) to the API service **106**. The verification request **140**(**1**) may include the TBTT **118**(**1**) or an identifier of the token information **130** associated with the TBTT **118**(**1**). In response to receipt of the verification request **140**(**1**), the API service **106** may determine a security context of the TBTT **118**(**1**) based upon the token information **130** stored during the generation of the TBTT **118**(**1**). Further, the security context may indicate the degree to which the TBTT **118**(**1**) has depreciated between issuance of the TBTT **118**(**1**) to the partner service **112**(**1**) and redemption of the TBTT **118**(**1**) by the client device **108**(**1**). In some embodiments, the API service **106** may employ a depreciation function to determine the security context of the TBTT **118**(**1**). For instance the API service **106** may employ a time-based depreciation function that determines the security context (e.g., full trust, limited trust, no trust, etc.) based upon the duration of time between issuance of the TBTT **118**(**1**) to the partner service **112**(**1**) and redemption of the TBTT **118**(**1**) by the client device **108**(**1**) (i.e., the age of the TBTT **118**(**1**) at the time of redemption).

(21) For example, the API service **106** may determine that the TBTT **118**(**1**) has not depreciated in trust (i.e., full trust) based upon the duration of time between issuance of the TBTT **118**(**1**) and redemption of the TBTT **118**(**1**) being less than a first predetermined value (e.g. fifteen minutes). As another example, the API service **106** may determine that the TBTT **118**(**1**) has depreciated to a lower level of trust (i.e., limited trust) based upon the duration of time between issuance of the TBTT **118**(**1**) and redemption of the TBTT **118**(**1**) being greater than the first predetermined value (e.g. fifteen minutes), and lower than a second predetermined value (e.g. thirty minutes). As yet still another example, the API service **106** may determine that the TBTT **118**(**1**) has depreciated to the lowest level of trust (i.e., no trust) based upon the duration of time between issuance of the TBTT **118**(**1**) and redemption of the TBTT **118**(**1**) being greater than a second predetermined value. Once the API service **106** determines the security context of the TBTT **118**(**1**), the API service **106** may send a verification response **142**(**1**) to the service provider **102**. The verification response **142**(**1**) may identify the security context of the TBTT **118**(**1**). Although three trust levels are discussed herein, the security context of the TBTT **118**(**1**) may be mapped to less than three trust levels or more than three trust levels.

(22) Further, in some embodiments, the depreciation function of the API service **106** may determine the security context of the TBTT **118**(**1**) based at least in part on non-temporal information. For example, the API service **106** may further determine the security context of the

TBTT **118**(**1**) based upon usage history of the clients **110**(**1**)-(N), client preferences, location information associated with the client devices **108**(**1**)-(N), attributes of web browsers used to login to the service provider **102**, or one or more device attributes of the client devices **108**(**1**)-(N). For example, the API service **106** may determine that a trust level of the TBTT **118**(**1**) has depreciated based upon the TBTT **118**(**1**) being received from a web browser that has not been previously used by the client device **108**(**1**) or the client **110**(**1**). As another example, the API service **106** may determine that a trust level of the TBTT **118**(**1**) has depreciated based upon the client device **108**(**1**) being located in a geographic location not previously associated with the client device **108**(**1**) or the client **110**(**1**). As yet still another example, the API service **106** may determine that a trust level of the TBTT **118**(**1**) has depreciated based upon the client device **108**(**1**) being an electronic device not previously used by the client **110**(**1**) with the service provider **102** or API service **106**.
(23) As further described below with respect to FIG. **2**, if the API service **106** determines that the security context of the TBTT **118**(**1**) is full trust, the service provider **102** may create the connection between the partner service **112**(**1**) and a client account of the client **110**(**1**). In some embodiments, the service provider **102** may employ a method call of the API service **106** to create the connection. Further, the service provider **102** or the API service **106** may send a connection notification **144**(**1**) to the client device **108**(**1**) and/or the partner service **112**(**1**). The connection notification **144**(**1**) may indicate that the connection has been created between a client account of the client **110**(**1**) and the partner service **112**(**1**). Upon creation of the connection, the partner service **112**(**1**) may assign the service tasks **120**(**1**)-(N) to the client **110**(**1**). For instance, the partner service **112**(**1**) generate the service task **120**(**1**) describing a bill owed by the client **110**(**1**) to the partner service **112**(**1**), and assign the service task **120**(**1**) to the client **110**(**1**) via the task management application **104** of the service provider **102**.
(24) Further, the service provider **102** may send the service task **120**(**1**) to the client device **108**(**1**) based on the connection. Upon receipt of the service task **120**(**1**), the client device **108**(**1**) may display the service task **120**(**1**) via a graphical user interface (GUI) **146**. In addition, the task management application **104** may present one or more actions that the task management application **104** can perform in view of the service task **120**(**1**).
(25) For instance, the task management application **104** may present a control **148** (e.g., a button) for paying the bill associated with the service task **120**(**1**). In response to the client **110**(**1**) selecting the control **148**, the task management application **104** may perform an action. For example, the task management application **104** may send a payment request to the service provider **102**, and the service provider **102** may facilitate a funds transfer from a financial account of the client **110**(**1**) to a financial account associated with the partner service **112**(**1**) based on the payment request.
(26) As further described below with respect to FIG. **3**, if the API service **106** determines that the security context of the TBTT **118**(**1**) is limited trust, the service provider **102** may require the client **110**(**1**) to perform additional authentication steps before creating the connection between the client **110**(**1**) and the partner service **112**(**1**). For example, the service provider **102** may request demographic information **150**(**1**) from the client device **108**(**1**). In some embodiments, the service provider **102** may request the demographic information **150**(**1**) based upon an attribute of the partner service **112**(**1**). For example, if the partner service **112**(**1**) is a cellular telephone service provider, the demographic attribute **150**(**1**) may be related to cellular phone use by the client, e.g., a telephone number or of the client. In some other embodiments, the service provider **102** may prompt for demographic information **150**(**1**) matching the demographic attributes **126**.
(27) Upon receipt of the demographic information **150**(**1**), the API service **106** may compare the demographic information **150**(**1**) to the newly acquired demographic information **152**(**1**) received from the partner service **112**(**1**) or the demographic attributes **126** previously received from the partner service **112**(**1**). If the demographic information **150**(**1**) matches the demographic information **152**(**1**) or the demographic attributes **126**, the service provider **102** may create the connection between a client account of the client **110**(**1**) and the partner service **112**(**1**). As such, the

service provider **102** may utilize the security context and connection information of a depreciated token (i.e., the TBTT **118**(**1**)) to create the connection instead of completely disregarding the TBTT **118**(**1**) once it no longer has a full trust security context.
(28) Further, the service provider **102** or the API service **106** may send the connection notification **144**(**1**) to the client device **108**(**1**) and/or the partner service **112**(**1**). Upon creation of the connection, the partner service **112**(**1**) may assign tasks to the client **110**(**1**). For instance, as described in detail above, the partner service **112**(**1**) may send the service task **120**(**1**) to the client device **108**(**1**) via the connection, the task management application **104** may display the service task **120**(**1**) to the client **110**(**1**) via the client device **108**(**1**), and the task management application **104** may facilitate performance of an action associated with the service task **120**(**1**) based on user selection of the control **148** presented within the GUI **146**.
(29) As further described below with respect to FIG. **4**, if the API service **106** determines that the security context of the TBTT **118**(**1**) is no trust and reject the connection request **124**(**1**). Further, the API service **106** may identify partner services **112**(**1**)-(N) associated with the client **110**(**1**), and the service provider **102** may present the identified partner services **112**(**1**)-(N) to the client **110**(**1**) via the task management application **104**. Further, the service provider **102** and API service **106** may facilitate the creation of a connection between the client **110**(**1**) and the identified partner services **112**(**1**)-(N).
(30) For example, the API service **106** may identify client information **128**(**1**) associated with the client **110**(**1**). In some embodiments, the client information **128**(**1**) may differ from the demographic attributes **126** received from partner service **112**(**1**). For instance, the client information **128**(**1**) may be user information maintained by the API service **106** or the service provider **102** with respect to the client **110**(**1**). Further, the API service **106** may send the client information **128**(**1**) to the partner platform **114**. The partner platform **114** may identify a plurality of partner services **112**(**1**)-(N) associated with the client **110**(**1**) based upon the client information **128**(**1**).
(31) For instance, the partner platform **114** may have access to a data store including client profiles **154**(**1**)-(N) for the clients **110**(**1**)-(N). In some examples, the client profiles **154**(**1**)-(N) may identify relationships between the clients **110**(**1**)-(N) and the partner services **112**(**1**)-(N). As an example, the client profile **154**(**1**) for the client **110**(**1**) may identify that the client **110**(**1**) has a relationship with the partner services **112**(**1**)-(**2**). Further, the partner platform **114** may search the client profile **154**(**1**) using the client information **128**(**1**) to determine that the client **110**(**1**) has a pre-existing relationship with the partner services **112**(**1**)-(**2**). Once the partner platform **114** has identified the relationship between the client **110**(**1**) and the partner services **112**(**1**)-(**2**), the partner platform **114** may send the API service **106** identifiers identifying the partner services **112**(**1**)-(**2**). Additionally, the API service **106** may instruct the service provider **102** to present the partner services **112**(**1**)-(**2**) to the client **110**(**1**) via the task management service **104**. In some embodiments, the task management service **104** may maintain state information indicating that the partner service **112**(**1**) is associated with the TBTT **118**(**1**) redeemed by the client **110**(**1**).
(32) Further, as an example, the client **110**(**1**) may request creation of a connection with the partner services **112**(**1**) via the task management service **104**. In response to a request to create a connection with the partner service **112**(**2**), the service provider **102** may prompt the client **110**(**1**) for a challenge response, and request the expected response to the challenge response from the partner service **112**(**2**). In addition, the API service **106** may compare the challenge response to the expected response. If the challenge response matches the expected response, the API service **106** may create the connection between the client **110**(**1**) and the partner service **112**(**1**). Upon creation of the connection, the partner service **112**(**1**) may assign the service tasks **120**(**1**)-(N) to the client **110**(**1**). For instance, as described in detail above, the partner service **112**(**1**) may send the service task **120**(**1**) to the client device **108**(**1**) via the connection, the task management application **104** may display the service task **120**(**1**) to the client **110**(**1**) via the client device **108**(**1**), and the task

management application **104** may facilitate performance of an action associated with the service task **120**(**1**) based on user selection of the control **148** presented within the GUI **146**.

(33) FIG. **2** is a sequence diagram illustrating an example process using time-based token trust depreciation, according to embodiments of the present disclosure. At step **202**, a client **110**(**1**) may visit the partner service **112**(**1**) or partner platform **114** with the intent of establishing a connection with the partner service **112**(**1**) at the service provider **102**. For example, the client **110**(**1**) may have a pre-existing relationship with the partner service **112**(**1**), and endeavor to manage tasks related to the partner service **112**(**1**) via the service provider **102**.

(34) At step **204**, the client **110**(**1**) may click an enrollment button within a graphical user interface presented by the partner service **112**(**1**) or partner platform **114** to the client device **108**(**1**), thereby consenting to the creation of a connection between the client **110**(**1**) and the partner service **112**(**1**) via the service provider **102**. Further, the client device **108**(**1**) may send the enrollment information **122**(**1**) to the partner service **112**(**1**) or the partner platform **114**.

(35) At step **206**, the partner service **112**(**1**) or partner platform **114** may send the connection request **124**(**1**) to the API service **106**. In some embodiments, the connection request **124**(**1**) may include one or more demographic attributes **126** associated with client **110**(**1**). Some examples of a demographic attributes include a name, email address, telephone number, physical address associated with the client **110**(**1**). Further, in some instances, the connect request **124**(**1**) may include a client identifier identifying the client **110**(**1**) and a partner service identifier identifying the partner service **112**(**1**).

(36) At step **208**, the API service **106** may receive the connection request **124**, generate the TBTT **118**(**1**) based on the connection request **124**, and send the connection information **134**(**1**) to the partner service **112**(**1**). The connection information **134**(**1**) may include the connection request identifier **132**(**1**) and the web application identifier **136**(**1**). In some embodiments, the web application identifier **136**(**1**) may be a URL link including the TBTT **118**(**1**). Further, the URL link may be the address of a login page of the service partner **102**.

(37) At step **210**, the partner service **112**(**1**) may redirect the client device **108**(**1**) to the login page associated with the URL link of the web application identifier **136**(**1**). As used herein, in some embodiments, a “redirect” may refer to a technique for forwarding a web visitor from a first domain to a second domain associated with a different URL. For instance, at step **210**, the client device **108**(**1**) may initially visit the partner service **112**(**1**) via a web browser, and the domain of the partner service **112**(**1**) may cause the domain of the service provider **102** to load within the web browser.

(38) At step **212**, the client device **108**(**1**) may login into the service provider **102** at the URL link. For example, the client device **110**(**1**) may load a webpage of the service provider **102**, and provide the login information **138**(**1**) to the service provider **102**. In some embodiments, the login information **138**(**1**) may include authentication credentials and the TBTT **118**(**1**). Some examples of authentication credentials include a password, a PIN code, a digital certificate, a secret, a social security number, or biometric information.

(39) At step **214**, the service provider **102** may send a verification request **140**(**1**) including the TBTT **118**(**1**) to the API service **106**. Further, upon receipt of the verification request **140**(**1**), the API service **106** may determine the security context of the TBTT **118**(**1**) based on a depreciation function and the TBTT **118**(**1**). For example, the API service **106** may determine that the security context of the TBTT **118**(**1**) is full trust based upon the elapsed time between issuance of the TBTT **118**(**1**) and redemption of the TBTT **118**(**1**) being less than a predetermined value.

(40) At step **216**, the API service **106** may send a verification response **142**(**1**) to the service provider **102** indicating that the service provider **102** should create the connection between the client **110**(**1**) and the partner service **112**(**1**) based on the security context of the TBTT **118**(**1**) being full trust. As described in detail herein, once the connection has been created between the client **110**(**1**) and the partner service **112**(**1**), the partner service **112**(**1**) may send the service task **120**(**1**)

to the client device **110**(**1**) via the task management application **104**. Further, the service provider **102** or API service **106** may send the connection notification **144** to the client device **108**(**1**) or the partner service **112**(**1**) indicating that the connection has been created.
(41) FIG. **3** is a sequence diagram illustrating an example process using time-based token trust depreciation, according to embodiments of the present disclosure. At step **302**, a client **110**(**1**) may visit the partner service **112**(**1**) or partner platform **114** with the intent of establishing a connection with the partner service **112**(**1**) at the service provider **102**. For example, the client **110**(**1**) may have a pre-existing relationship with the partner service **112**(**1**), and endeavor to manage tasks related to the partner service **112**(**1**) via the service provider **102**.
(42) At step **304**, the client **110**(**1**) may click an enrollment button within a graphical user interface presented by the partner service **112**(**1**) or partner platform **114** to the client device **108**(**1**), thereby consenting to the creation of a connection between the client **110**(**1**) and the partner service **112**(**1**) via the service provider **102**. Further, the client device **108**(**1**) may send the enrollment information **122**(**1**) to the partner service **112**(**1**) or the partner platform **114**.
(43) At step **306**, the partner service **112**(**1**) or partner platform **114** may send the connection request **124**(**1**) to the API service **106**. In some embodiments, the connection request **124**(**1**) may include one or more demographic attributes **126** associated with client **110**(**1**). Some examples of a demographic attributes include a name, email address, telephone number, physical address associated with the client **110**(**1**). Further, in some instances, the connect request **124**(**1**) may include a client identifier identifying the client **110**(**1**) and a partner service identifier identifying the partner service **112**(**1**).
(44) At step **308**, the API service **106** may receive the connection request **124**, generate the TBTT **118**(**1**) based on the connection request **124**, and send the connection information **134**(**1**) to the partner service **112**(**1**). The connection information **134**(**1**) may include the connection request identifier **132**(**1**) and the web application identifier **136**(**1**). In some embodiments, the web application identifier **136**(**1**) may be a URL link including a TBTT **118**(**1**). Further, the URL link may be the address of a login page of the service partner **102**.
(45) At step **310**, the partner service **112**(**1**) may redirect the client device **108**(**1**) to the login page associated with the URL link of the web application identifier **136**(**1**). As used herein, in some embodiments, a “redirect” may refer to a technique for forwarding a web visitor from a first domain to a second domain associated with a different URL. For instance, at step **210**, the client device **108**(**1**) may initially visit the partner service **112**(**1**) via a web browser, and the domain of the partner service **112**(**1**) may cause the domain of the service provider **102** to load within the web browser.
(46) At step **312**, the client device **110**(**1**) may login into the service provider **102** at the URL link. For example, the client device **110**(**1**) may load a webpage of the service provider **102**, and provide the login information **138**(**1**) to the service provider **102**. In some embodiments, the login information **138**(**1**) may include authentication credentials and the TBTT **118**(**1**). Some examples of authentication credentials include a password, a PIN code, a digital certificate, a secret, a social security number, or biometric information.
(47) At step **314**, the service provider **102** may send a verification request **140**(**1**) including the TBTT **118**(**1**) to the API service **106**. Further, upon receipt of the verification request **140**(**1**), the API service **106** may determine the security context of the TBTT **118**(**1**) based on a depreciation function and the TBTT **118**(**1**). For example, the API service **106** may determine that the security context of the TBTT **118**(**1**) is low trust based upon the elapsed time between issuance of the TBTT **118**(**1**) and redemption of the TBTT **118**(**1**) being greater than a first predetermined value, and lesser than a second predetermined value.
(48) At step **316**, the API service **106** may send the verification response **142**(**1**) to the service provider **102**. Further, the verification response **142**(**1**) may instruct the service provider **102** to request the demographic information **150**(**1**) from the client device **110**(**1**) based on the security

context of the TBTT **118**(**1**). In some embodiments, the API service **106** may instruct the service provider **102** to request one demographic attribute. In some other embodiments, the API service **106** may instruct the service provider **102** to request a plurality of demographic attributes. Further, the amount of demographic attributes requested by the service provider **102** may correspond to the depreciation of the TBTT **118**(**1**). For instance, if the TBTT **118**(**1**) has depreciated multiple levels of trust, the service provider **102** may request a plurality of demographic attributes from the client device **110**(**1**) to verify the identity of the client **110**(**1**).

(49) At step **318**, the service provider **102** may request the demographic information **150**(**1**) from the client device **110**(**1**). For example, the service provider **102** may prompt the client device **108**(**1**) for the date of birth of the client **110**(**1**). At step **320**, the service provider **102** may receive the demographic information **150**(**1**) in response to the prompt. In some embodiments, the service provider **102** may also request the answer to the prompt from the partner service **112**(**1**). In some other embodiments, the service provider **102** or API service **106** may locally store the expected response (i.e., the demographic attributes **126** or the client information **128**) to the prompt.

(50) At step **322**, the API service **106** may receive a verification request **140**(**2**) including the demographic information from the service provider **102**, and verify the demographic information **150**(**1**) by comparing the demographic information to the expected response. At step **324**, the API service **106** may send the verification response **142**(**1**) to the service provider **102**. If the demographic information **150**(**1**) matches the expected response, the verification response **142**(**1**) may indicate that the service provider **102** should create the connection between the client **110**(**1**) and the partner service **112**(**1**). Otherwise, the verification response **142**(**1**) may indicate that the service provider **102** should not create the connection between the client **110**(**1**) and the partner service **112**(**1**). As described in detail herein, once the connection has been created between the client **110**(**1**) and the partner service **112**(**1**), the partner service **112**(**1**) may send the service task **120**(**1**) to the client device **110**(**1**) via the task management application **104**. Further, the service provider **102** or API service **106** may send the connection notification **144** to the client device **108**(**1**) or the partner service **112**(**1**) indicating that the connection has been created.

(51) FIG. **4** is a sequence diagram illustrating an example process using time-based token trust depreciation, according to embodiments of the present disclosure. At step **402**, a client **110**(**1**) may visit the partner service **112**(**1**) or partner platform **114** with the intent of establishing a connection with the partner service **112**(**1**) at the service provider **102**. For example, the client **110**(**1**) may have a pre-existing relationship with the partner service **112**(**1**), and endeavor to manage tasks related to the partner service **112**(**1**) via the service provider **102**.

(52) At step **404**, the client **110**(**1**) may click an enrollment button within a graphical user interface presented by the partner service **112**(**1**) or partner platform to the client device **108**(**1**), thereby consenting to the creation of a connection between the client **110**(**1**) and the partner service **112**(**1**) via the service provider **102**. Further, the client device **108**(**1**) may send the enrollment information **122**(**1**) to the partner service **112**(**1**) or the partner platform **114**.

(53) At step **406**, the partner service **112**(**1**) or partner platform **114** may send the connection request **124**(**1**) to the API service **106**. In some embodiments, the connection request **124**(**1**) may include one or more demographic attributes **126** associated with client **110**(**1**). Some examples of a demographic attributes include a name, email address, telephone number, physical address associated with the client **110**(**1**). Further, in some instances, the connect request **124**(**1**) may include a client identifier identifying the client **110**(**1**) and a partner service identifier identifying the partner service **112**(**1**).

(54) At step **408**, the API service **106** may receive the connection request **124**, generate the TBTT **118**(**1**) based on the connection request **124**, and send the connection information **134**(**1**) to the partner service **112**(**1**). The connection information **134**(**1**) may include the connection request identifier **132**(**1**) and the web application identifier **136**(**1**). In some embodiments, the web application identifier **136**(**1**) may be a URL link including a TBTT **118**(**1**). Further, the URL link

may be the address of a login page of the service partner **102**.

(55) At step **410**, the partner service **112**(**1**) may redirect the client device **108**(**1**) to the login page associated with the URL link of the web application identifier **136**(**1**). As used herein, in some embodiments, a "redirect" may refer to a technique for forwarding a web visitor from a first domain to a second domain associated with a different URL. For instance, at step **210**, the client device **108**(**1**) may initially visit the partner service **112**(**1**) via a web browser, and the domain of the partner service **112**(**1**) may cause the domain of the service provider **102** to load within the web browser.

(56) At step **412**, the client device **110**(**1**) may login into the service provider **102** at the URL link. For example, the client device **110**(**1**) may load a webpage of the service provider **102**, and provide the login information **138**(**1**) to the service provider **102**. In some embodiments, the login information **138**(**1**) may include authentication credentials and the TBTT **118**(**1**). Some examples of authentication credentials include a password, a PIN code, a digital certificate, a secret, a social security number, or biometric information.

(57) At step **414**, the service provider **102** may send a verification request **140**(**1**) including the TBTT **118**(**1**) to the API service **106**. Further, upon receipt of the verification request **140**(**1**), the API service **106** may determine the security context of the TBTT **118**(**1**) based on a depreciation function and the TBTT **118**(**1**). For example, the API service **106** may determine that the security context of the TBTT **118**(**1**) is no trust based upon the elapsed time between issuance of the TBTT **118**(**1**) and redemption of the TBTT **118**(**1**) being greater than a predetermined value.

(58) At step **416**, the API service **106** may request a list of the partner services **112**(**1**)-(N) affiliated with the client **110**(**1**) from the partner platform **114** based on the security context of the TBTT **118**(**1**). In some embodiments, the request may include the client information **128**.

(59) At step **418**, the partner platform **114** may receive the client information **128**, determine that the partner services **112**(**1**)-(**2**) are associated with the client **110**(**1**) based upon the client information **128**, and send a list including the partner services **112**(**1**)-(**2**) to the API service **106**. For instance, the partner platform **114** may search the client profile **152**(**1**) using the client information **128** to determine that the client **110**(**1**) has a pre-existing relationship with the partner services **112**(**1**)-(**2**). Once the partner platform **114** has identified the relationships between the client **110**(**1**) and the partner services **112**(**1**)-(**2**), the partner platform **114** may send partner identifiers identifying the partner services **112**(**1**)-(**2**) to the API service **106**.

(60) At step **420**, the API service **106** may instruct the service partner **102** to present the partner services **112**(**1**)-(**2**) to the client **110**(**1**) via the task management service **104**. At step **422**, the service provider **102** may present the partner services **112**(**1**)-(**2**) within the task management application **104**. Further, the client **110**(**1**) may select to create connections with the partner services **112**(**1**)-(**2**) within the task management application **104**. In some embodiments, the API service **106** may perform additional authentication steps before creating the connections between the client **110**(**1**) and the partner services **112**(**1**)-(**2**), respectively.

(61) FIG. **5** is a flowchart illustrating a process for implementing location identification in MFA, according to some embodiments. Method **500** can be performed by processing logic that can comprise hardware (e.g., circuitry, dedicated logic, programmable logic, microcode, etc.), software (e.g., instructions executing on a processing device), or a combination thereof. It is to be appreciated that not all steps may be needed in all embodiments. Further, some of the steps may be performed simultaneously, or in a different order than shown in FIG. **4**, as will be understood by a person of ordinary skill in the art.

(62) Method **500** shall be described with reference to FIG. **1**. However, method **500** is not limited to those example embodiments.

(63) In **502**, a system may receive a connection request from a partner device having obtained consent to connect to a client account within a secure platform, the connection request including a demographic attribute associated with a first client account. For example, the partner service **112**(**1**)

may send the connection request **124**(**1**) to the API service **106** in response to receipt of the enrollment information **122**(**1**) from the client **110**(**1**). In some embodiments, the connection request **124**(**1**) may include one or more demographic attributes **126** associated with client **110**(**1**). Some examples of a demographic attributes **126** include a name, email address, telephone number, physical address associated with the client **110**(**1**).
(64) In **504**, the system may match the demographic attribute to client information associated with the first client account. For example, upon receipt of the connection request **124**(**1**), the API service **106** may determine whether the demographic attribute **126** matches client information **128** associated with the client **110**(**1**). If the demographic attribute **126** matches the client information **128**, the API service **106** may generate the TBTT **118**(**1**) for creating the connection between the client **110**(**1**) and the partner service **112**(**1**) within the task management application **104**.
(65) In **506**, the system may send, to the partner device, an identifier and a URL-link including a depreciating token, the identifier identifying the connection request and the URL-link configured to redirect a client device to a login page. For example, the API service **106** may send the connection information **134**(**1**) to the partner service **112**(**1**). The connection information **134**(**1**) may include a web application identifier **136**(**1**) including the TBTT **118**, and the connection request identifier **132**(**1**). In some examples, the web application identifier **136**(**1**) may be a URL-link including the TBTT **118** as a query string.
(66) In **508**, the system may authenticate a second client account via the login page. For example, the client **110**(**1**) may perform successfully authenticate to the service provider **102** via the login page associated with the web application identifier **136**(**1**). In some embodiments, the client device **108**(**1**) may send the login information **138** including the TBTT **118**(**1**) to the service provider **102** during the authentication process.
(67) In **510**, the system may receive a verification request including the identifier and the depreciating token. For example, the service provider **102** may send a verification request **140**(**1**) to the API service **106**. The verification request **140**(**1**) may include the TBTT **118**(**1**) or an identifier of the token information **130** associated with the TB TT **118**(**1**).
(68) In **512**, the system may determine a security context of the depreciating token based on a depreciation function and the verification request. In response to receipt of the verification request **140**(**1**), the API service **106** may determine a security context of the TBTT **118**(**1**) based a depreciation function. In some embodiments, the API service **106** may employ a time-based depreciation function that determines the security context (e.g., full trust, limited trust, no trust, etc.) based upon the duration of time between issuance of the TBTT **118**(**1**) to the partner service **112**(**1**), and redemption of the TBTT **118**(**1**) by the client device **108**(**1**).
(69) In **514**, the system may determine, based on the security context, whether to create a connection between the second client account and partner device within the secure platform. Once the API service **106** determines the security context of the TBTT **118**(**1**), the API service **106** may send a verification response **142**(**1**) to the service provider **102**. The verification response **142**(**1**) may identify the security context of the TBTT **118**(**1**), or otherwise indicate whether the service provider should create a connection between the client **110**(**1**) and the partner service **112**(**1**). If the security context is full trust, the verification response **142**(**1**) may indicate that the service provider should create the connection between the client **110**(**1**) and the partner service **112**(**1**).
(70) As described in detail above, the partner service **112**(**1**) may send the service task **120**(**1**) to the client device **108**(**1**) via the connection. Further, the task management application **104** may display the service task **120**(**1**) to the client **110**(**1**) via the client device **108**(**1**), and the task management application **104** may facilitate performance of an action associated with the service task **120**(**1**) based on user selection of the control **148** presented within the GUI **146**.
(71) FIG. **6** is a block diagram of example components of system **600**. To implement one or more example embodiments, one or more systems **600** may be used, for example, to implement embodiments discussed herein, as well as combinations and sub-combinations thereof. System **600**

may include one or more processors (also called central processing units, or CPUs), such as a processor **604**. Processor **604** may be connected to a communication infrastructure or bus **606**.
(72) System **600** may also include user input/output device(s) **603**, such as monitors, keyboards, pointing devices, etc., which may communicate with communication infrastructure **606** through user input/output interface(s) **602**.
(73) One or more of processors **604** may be a graphics processing unit (GPU). In an embodiment, a GPU may be a processor that is a specialized electronic circuit designed to process mathematically intensive applications. The GPU may have a parallel structure that is efficient for parallel processing of large blocks of data, such as mathematically intensive data common to computer graphics applications, images, videos, etc.
(74) System **600** may also include a main or primary memory **608**, such as random access memory (RAM). Main memory **608** may include one or more levels of cache. Main memory **608** may have stored therein control logic (i.e., computer software) and/or data.
(75) System **600** may also include one or more secondary storage devices or memory **610**. Secondary memory **610** may include, for example, a hard disk drive **612** and/or a removable storage device or drive **614**.
(76) Removable storage drive **614** may interact with a removable storage unit **618**. Removable storage unit **618** may include a computer usable or readable storage device having stored thereon computer software (control logic) and/or data. Removable storage unit **618** may be program cartridge and cartridge interface (such as that found in video game devices), a removable memory chip (such as an EPROM or PROM) and associated socket, a memory stick and USB port, a memory card and associated memory card slot, and/or any other removable storage unit and associated interface. Removable storage drive **614** may read from and/or write to removable storage unit **618**.
(77) Secondary memory **610** may include other means, devices, components, instrumentalities or other approaches for allowing computer programs and/or other instructions and/or data to be accessed by system **600**. Such means, devices, components, instrumentalities or other approaches may include, for example, a removable storage unit **622** and an interface **620**. Examples of the removable storage unit **622** and the interface **620** may include a program cartridge and cartridge interface (such as that found in video game devices), a removable memory chip (such as an EPROM or PROM) and associated socket, a memory stick and USB port, a memory card and associated memory card slot, and/or any other removable storage unit and associated interface.
(78) System **600** may further include a communication or network interface **624**. Communication interface **624** may enable system **600** to communicate and interact with any combination of external devices, external networks, external entities, etc. (individually and collectively referenced by reference number **628**). For example, communication interface **624** may allow system **600** to communicate with external or remote devices **628** over communications path **626**, which may be wired and/or wireless (or a combination thereof), and which may include any combination of LANs, WANs, the Internet, etc. Control logic and/or data may be transmitted to and from system **600** via communication path **626**.
(79) System **600** may be a client or server, accessing or hosting any applications and/or data through any delivery paradigm, including but not limited to remote or distributed cloud computing solutions; local or on-premises software ("on-premise" cloud-based solutions); "as a service" models (e.g., content as a service (CaaS), digital content as a service (DCaaS), software as a service (SaaS), managed software as a service (MSaaS), platform as a service (PaaS), framework as a service (FaaS), backend as a service (BaaS), mobile backend as a service (MBaaS), infrastructure as a service (IaaS), etc.); and/or a hybrid model including any combination of the foregoing examples or other services or delivery paradigms. In some embodiments, a tangible, non-transitory apparatus or article of manufacture comprising a tangible, non-transitory computer useable or readable medium having control logic (software) stored thereon may also be referred to herein as a

computer program product or program storage device. This includes, but is not limited to, system **600**, main memory **608**, secondary memory **610**, and removable storage units **618** and **622**, as well as tangible articles of manufacture embodying any combination of the foregoing. Such control logic, when executed by one or more data processing devices (such as computer system **600**), may cause such data processing devices to operate as described herein.
(80) It is to be appreciated that the Detailed Description section, and not Abstract section, is intended to be used to interpret the claims. The Abstract section may set forth one or more but not all example embodiments of the present invention as contemplated by the inventor(s), and thus, are not intended to limit the present invention and the appended claims in any way.
(81) The present invention has been described above with the aid of functional building blocks illustrating the implementation of specified functions and relationships thereof. The boundaries of these functional building blocks have been arbitrarily defined herein for the convenience of the description. Alternate boundaries can be defined so long as the specified functions and relationships thereof are appropriately performed.
(82) The foregoing description of the specific embodiments will so fully reveal the general nature of the invention that others can, by applying knowledge within the skill of the art, readily modify and/or adapt for various applications such specific embodiments, without undue experimentation, without departing from the general concept of the present invention. Therefore, such adaptations and modifications are intended to be within the meaning and range of equivalents of the disclosed embodiments, based on the teaching and guidance presented herein. It is to be understood that the phraseology or terminology herein is for the purpose of description and not of limitation, such that the terminology or phraseology of the present specification is to be interpreted by the skilled artisan in light of the teachings and guidance.
(83) Identifiers, such as “(a),” “(b),” “(i),” “(ii),” etc., are sometimes used for different elements or steps. These identifiers are used for clarity and do not necessarily designate an order for the elements or steps.
(84) The breadth and scope of the present invention should not be limited by any of the above-described example embodiments, but should be defined only in accordance with the following claims and their equivalents.

Claims

1. A method comprising: receiving, by one or more computing devices and from a client device, a connection request to a service, wherein the connection request includes one or more demographic attributes of a client; transmitting the connection request to an API service; receiving connection information from the API service based on the connection request, wherein the connection information includes a connection request identifier and a web application identifier, and wherein the web application identifier is a URL link including a depreciation token; redirecting the client device to a login page associated with the URL link; determining whether a connection with the client device should be established based on a verification process that determines whether a security context of the depreciation token is trusted based upon an elapsed time between issuance of the depreciation token and redemption of the depreciation token being less than a predetermined value; and based on establishing the connection, transmitting a service task to the client device upon establishing the connection.
2. The method of claim 1, wherein the connection request further includes a client identifier identifying the client and a partner service identifier identifying the service.
3. The method of claim 1, wherein the service task comprises providing a signature by a user to facilitate performance of an action.
4. The method of claim 1, wherein the service task comprises providing information by a user to facilitate performance of an action.

5. The method of claim 1, further comprising determining whether to not establish a connection with the client device based on the verification process determining whether the security context of the depreciation token is not trusted based upon the elapsed time between the issuance of the depreciation token and redemption of the depreciation token being equal to or greater than the predetermined value.
6. The method of claim 1, wherein the connection with the client device is established via a service provider intermediary.
7. The method of claim 1, further comprising transmitting the service task for display on a graphical user interface (GUI) of the client device.
8. A non-transitory computer-readable device having instructions stored thereon that, when executed by one or more processors of a computing system, cause the one or more processors to perform operations comprising: receiving, by one or more computing devices and from a client device, a connection request to a service, wherein the connection request includes one or more demographic attributes of a client; transmitting the connection request to an API service; receiving connection information from the API service based on the connection request, wherein the connection information includes a connection request identifier and a web application identifier, and wherein the web application identifier is a URL link including a depreciation token; redirecting the client device to a login page associated with the URL link; determining whether a connection with the client device should be established based on a verification process that determines whether a security context of the depreciation token is trusted based upon an elapsed time between issuance of the depreciation token and redemption of the depreciation token being less than a predetermined value; and transmitting a service task to the client device upon establishing the connection.
9. The non-transitory computer readable medium of claim 8, wherein the connection request further includes a client identifier identifying the client and a partner service identifier identifying the service.
10. The non-transitory computer readable medium of claim 8, wherein the service task comprises providing a signature by a user to facilitate performance of an action.
11. The non-transitory computer readable medium of claim 8, wherein the service task comprises providing information by a user to facilitate performance of an action.
12. The non-transitory computer readable medium of claim 8, wherein the operations further comprise determining whether to not establish a connection with the client device based on the verification process determining whether the security context of the depreciation token is not trusted based upon the elapsed time between the issuance of the depreciation token and redemption of the depreciation token being equal to or greater than the predetermined value.
13. The non-transitory computer readable medium of claim 8, wherein the connection with the client device is established via a service provider intermediary.
14. The non-transitory computer readable medium of claim 8, wherein the operations further comprising transmitting the service task for display on a graphical user interface (GUI) of the client device.
15. A computing system comprising: a memory storing instructions; one or more processors, coupled to the memory, configured to process the stored instructions to: receive, from a client device, a connection request to a service, wherein the connection request includes one or more demographic attributes of a client; transmit the connection request to an API service; receive connection information from the API service based on the connection request, wherein the connection information includes a connection request identifier and a web application identifier, and wherein the web application identifier is a URL link including a depreciation token; redirect the client device to a login page associated with the URL link; determine whether a connection with the client device should be established based on a verification process that determines whether a security context of the depreciation token is trusted based upon an elapsed time between issuance of the depreciation token and redemption of the depreciation token being less than a predetermined

value; and transmit a service task to the client device upon establishing the connection.
16. The computing system of claim 15, wherein the connection request further includes a client identifier identifying the client and a partner service identifier identifying the service.
17. The computing system of claim 15, wherein the service task comprises providing a signature by a user to facilitate performance of an action.
18. The computing system of claim 15, wherein the service task comprises providing information by a user to facilitate performance of an action.
19. The computing system of claim 15, wherein the one or more processors are further configured to determine whether to not establish a connection with the client device based on the verification process determining whether the security context of the depreciation token is not trusted based upon the elapsed time between the issuance of the depreciation token and redemption of the depreciation token being equal to or greater than the predetermined value.
20. The computing system of claim 15, wherein the one or more processors are further configured to transmit the service task for display on a graphical user interface (GUI) of the client device.